Deep Neural Video Surveillance for Smart Campus Safety

Department of Computer Science & Engineering • Major Project Phase II

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Abstract & Literature Survey

• Current campus CCTV networks rely on passive human monitoring, leading to delayed incident detection.
• Recent benchmarks (Redmon et al., 2023) show YOLOv8 achieves high inference speed but degrades under low-light occlusion.
• Identified Gap: Need for lightweight edge tensor inference with multi-target tracking without central server bandwidth bottlenecks.

System Architecture & Edge Pipeline

• Edge Ingestion: RTSP video streams captured at 1080p 30fps from 16 IP cameras.
• Detection & Tracking: Custom fine-tuned YOLOv10-Nano with ByteTrack associative tracker on Jetson Orin Nano.
• Backend Architecture: FastAPI WebSocket dispatcher pushing geo-tagged alert frames to local Redis cache.
• Database & Logging: PostgreSQL with TimescaleDB for spatial-temporal telemetry query acceleration.

Testing, Verification & Quantitative Results

- Mean Average Precision (mAP@0.50): 91.4% across 4,500 annotated campus perimeter frames.
- End-to-End Alert Latency: 142ms from camera sensor frame to security control-room alert pop-up.
- Concurrency Testing: Successfully sustained 16 concurrent RTSP streams at 28.5 avg FPS.
- Conclusion: 65% faster emergency response time verified during mock safety evacuation drill.